Database Management System(DBMS)

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Based on the View and Index

Use the following table structure for all questions.
Table: LibraryBook
`LibraryBook(BookID, BookTitle, ISBN, SupplierID, Price, Status)`

Section A: Views

1. Create the `LibraryBook` table with appropriate data types and constraints. Insert at least 8 records into the table.

```
mysql> desc LibraryBook;
+------------+---------------+------+-----+---------+-------+
| Field      | Type          | Null | Key | Default | Extra |
+------------+---------------+------+-----+---------+-------+
| BookID     | int           | NO   | PRI | NULL    |       |
| BookTitle  | varchar(100)  | NO   |     | NULL    |       |
| ISBN       | varchar(20)   | YES  | UNI | NULL    |       |
| SupplierID | int           | YES  |     | NULL    |       |
| Price      | decimal(10,2) | YES  |     | NULL    |       |
| Status     | varchar(20)   | YES  |     | NULL    |       |
+------------+---------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

2. Create a view named `AvailableBooks` to display only the `BookID`, `BookTitle`, `ISBN`, and `Price` of books whose status is `'Available'`.

```
mysql> CREATE VIEW AvailableBooks AS
    -> SELECT BookID, BookTitle, ISBN, Price
    -> FROM LibraryBook
    -> WHERE Status = 'Available';
Query OK, 0 rows affected (0.01 sec)
```

3. Display all records from the `AvailableBooks` view.

```
mysql> SELECT * FROM AvailableBooks;
+--------+------------------+--------+--------+
| BookID | BookTitle        | ISBN   | Price  |
+--------+------------------+--------+--------+
|      1 | DBMS             | 978101 | 450.00 |
|      3 | Python Basics    | 978103 | 550.00 |
|      5 | Web Development  | 978105 | 750.00 |
|      7 | Operating System | 978107 | 500.00 |
|      8 | Data Structures  | 978108 | 800.00 |
+--------+------------------+--------+--------+
5 rows in set (0.00 sec)
```

4. Create a view named `ExpensiveBooks` to display books with a price greater than ₹500.

```
mysql> create VIEW ExpensiveBooks as select BookID,BookTitle,ISBN,Price from LibraryBook where Price > 500;
Query OK, 0 rows affected (0.03 sec)
```

5. Display the book titles and prices from the `ExpensiveBooks` view in descending order of price.

```
mysql> select BookTitle,Price  from ExpensiveBooks ORDER BY Price DESC;
+--------------------+--------+
| BookTitle          | Price  |
+--------------------+--------+
| Data Structures    | 800.00 |
| Web Development    | 750.00 |
| Computer Networks  | 650.00 |
| Java Programming   | 600.00 |
| Python Basics      | 550.00 |
+--------------------+--------+
5 rows in set (0.00 sec)
```

6. Create a view named `BookSummary` to display the total number of books and the average price of books.

```
mysql> create VIEW BookSummary as select Count(BookID) as Total_Books ,AVG(Price) as Average_Price from LibraryBook;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> create VIEW BookSummary as select Count(BookID) as Total_Books ,AVG(Price) as Average_Price from LibraryBook;
Query OK, 0 rows affected (0.03 sec)

mysql> select * from BookSummary;
+-------------+---------------+
| Total_Books | Average_Price |
+-------------+---------------+
|           8 |    587.500000 |
+-------------+---------------+
1 row in set (0.00 sec)
```

7. Create a view named `BookStatusView` to display the book title and status of each book. Use this view to display only books that are currently issued.

```
mysql> create VIEW BookStatusView as select BookTitle,Status from LibraryBook where Status='Issued';
Query OK, 0 rows affected (0.00 sec)

mysql> select * from BookStatusView;
+-------------------+--------+
| BookTitle         | Status |
+-------------------+--------+
| Java Programming  | Issued |
| C Programming     | Issued |
| Computer Networks | Issued |
+-------------------+--------+
3 rows in set (0.00 sec)
```

8. Modify the `AvailableBooks` view so that it displays only the `BookID`, `BookTitle`, and `Price` columns.

```
mysql> Create OR Replace VIEW  AvailableBooks as select BookID,BookTitle,Price from LibraryBook;
Query OK, 0 rows affected (0.03 sec)

mysql> select * from AvalableBooks;
```

```
mysql> select * from AvailableBooks;
+--------+-------------------+--------+
| BookID | BookTitle         | Price  |
+--------+-------------------+--------+
|      1 | DBMS              | 450.00 |
|      2 | Java Programming  | 600.00 |
|      3 | Python Basics     | 550.00 |
|      4 | C Programming     | 400.00 |
|      5 | Web Development   | 750.00 |
|      6 | Computer Networks | 650.00 |
|      7 | Operating System  | 500.00 |
|      8 | Data Structures   | 800.00 |
+--------+-------------------+--------+
8 rows in set (0.00 sec)
```

9. Drop the `ExpensiveBooks` view.

```
mysql> Drop VIEW ExpensiveBooks;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from ExpensiveBooks;
ERROR 1146 (42S02): Table 'Harshad.ExpensiveBooks' doesn't exist
mysql>
```

Section B: Indexes

10. Create a normal index named `idx_booktitle` on the `BookTitle` column of the `LibraryBook` table.

```
mysql> Create INDEX idx_booktitle on LibraryBook(BookTitle);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

11. Display the indexes created on the `LibraryBook` table.

```
mysql> Show INDEX from LibraryBook;
+-------------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table       | Non_unique | Key_name     | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-------------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| LibraryBook |          0 | PRIMARY      |            1 | BookID      | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          0 | ISBN         |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_booktitle |            1 | BookTitle   | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
+-------------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
3 rows in set (0.00 sec)
```

12. Create an index named `idx_price` on the `Price` column.

```
mysql> create INDEX idx_price on LibraryBook(Price);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> Show INDEX from LibraryBook;
+-------------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+-----
| Table       | Non_unique | Key_name     | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Exp
+-------------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+-----
| LibraryBook |          0 | PRIMARY      |            1 | BookID      | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NUL
| LibraryBook |          0 | ISBN         |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NUL
| LibraryBook |          1 | idx_booktitle |            1 | BookTitle   | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NUL
| LibraryBook |          1 | idx_price    |            1 | Price       | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NUL
+-------------+------------+--------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+-----
4 rows in set (0.00 sec)
```

13. Create a composite index named `idx_status_price` on the `Status` and `Price` columns.

```
mysql> create INDEX  idx_status_price  on LibraryBook(Status,Price);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> Show INDEX from LibraryBook;
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table       | Non_unique | Key_name        | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| LibraryBook |          0 | PRIMARY         |            1 | BookID      | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          0 | ISBN            |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_booktitle   |            1 | BookTitle   | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_price       |            1 | Price       | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           1 | Status      | A         |           2 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           2 | Price       | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
```

14. Create a unique index named `idx_isbn` on the `ISBN` column to prevent duplicate ISBN values.

```
mysql> create INDEX idx_isbn on LibraryBook(ISBN);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> Show INDEX from LibraryBook;
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table       | Non_unique | Key_name        | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| LibraryBook |          0 | PRIMARY         |            1 | BookID      | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          0 | ISBN            |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_booktitle   |            1 | BookTitle   | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_price       |            1 | Price       | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           1 | Status      | A         |           2 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           2 | Price       | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_isbn        |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
```

15. Drop the `idx_price` index from the `LibraryBook` table.

```
mysql> Drop INDEX idx_price on LibraryBook;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> Show INDEX from LibraryBook;
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table       | Non_unique | Key_name        | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| LibraryBook |          0 | PRIMARY         |            1 | BookID      | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          0 | ISBN            |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_booktitle   |            1 | BookTitle   | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           1 | Status      | A         |           2 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           2 | Price       | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_isbn        |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
```

16. Compare the indexes displayed before and after dropping `idx_price`.

Before Dropping:

```
mysql> Show INDEX from LibraryBook;
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table       | Non_unique | Key_name        | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| LibraryBook |          0 | PRIMARY         |            1 | BookID      | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          0 | ISBN            |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_booktitle   |            1 | BookTitle   | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_price       |            1 | Price       | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           1 | Status      | A         |           2 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           2 | Price       | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_isbn        |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
```

After Dropping:

```
mysql> SHOW INDEX FROM LibraryBook;
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table       | Non_unique | Key_name        | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| LibraryBook |          0 | PRIMARY         |            1 | BookID      | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          0 | ISBN            |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_booktitle   |            1 | BookTitle   | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           1 | Status      | A         |           2 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_status_price |           2 | Price       | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| LibraryBook |          1 | idx_isbn        |            1 | ISBN        | A         |           8 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+-------------+------------+-----------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
```